

Shut the border and white America still shrinks

A population forecast for every American state to 2075, with odds, and what it says about the "great replacement"

The white share of America's population is falling, as the replacement theory says. Most of the fall would happen with no immigration at all, its size depends on who is counted as white, and in 20 states the white majority will probably outlast the forecast

Census Bureau population estimates 2000-2025 (Vintage 2025) and 2023 national projections; projection 2025-2075

92%

Chance that non-Hispanic white people are under half of Americans by 2050. They are 56% today

2038

Most likely year the share falls below half (80% range 2035-47)

36% or 71%

White share in 2075, counting only non-Hispanic people who are white alone, or everyone classed as white or part-white

20

States more likely than not to keep a white majority to 2075 on the narrowest count

In 2011 Renaud Camus, a French writer, gave a name to an old fear. "The great replacement" holds that white populations in Europe and America are being swapped for others, and that elites are arranging it. The phrase has since travelled from the fringe to cable news and campaign speeches.

The theory makes two claims. One is about numbers: the white share of the population is falling. The other is about intent: someone is doing it on purpose. No population data can test the second, and nothing in this report supports it. The first can be checked. This report projects America's population to 2075 for all 50 states and Washington, DC, by age, sex, race and Hispanic origin, and attaches odds to the results.

It does so in four steps. First, a standard cohort projection carries each state's population forward in five-year steps, and is tested on the past, back to 1990. Second, a Bayesian model of how much the projection's inputs have moved from year to year turns one projection into 4,000 simulated futures. Third, the same futures are read under three definitions of white. Fourth, the Census Bureau's own scenarios are used to separate ageing from immigration, which our state data cannot do; our own no-immigration run repeats the exercise on current data; the Bureau's survey shows who arrives in each state; and a what-if asks how many children it would take to change the result.

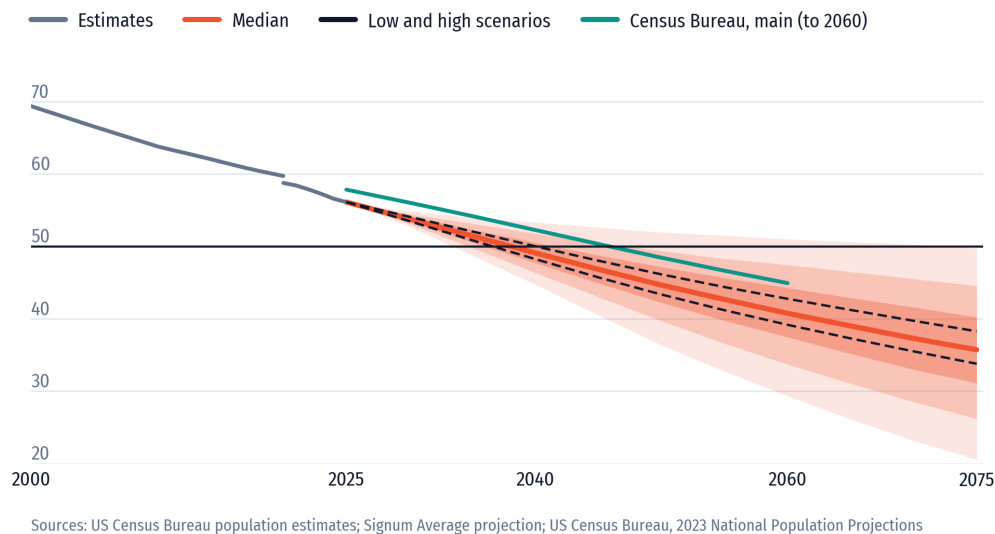
What the numbers say

In 2000 non-Hispanic white people were 69.4% of Americans, 196m of 282m. In 2025 they were 56.1%, 192m of 342m. Their number barely moved for two decades and has fallen by 3m since 2020; the rest of the population grew from 86m to 150m. (The Census Bureau changed the race base of its estimates in 2020, which lowers the white share by about one point; the comparison across that break is approximate.)

The median of our simulated futures puts the share at 49% in 2040, 45% in 2050 and 36% in 2075. The 80% range for 2075 runs from 26% to 45%, and the 95% range from 21% to 50%. In three futures out of four the share is below 40% in 2075; in one in five it is below 30%.

The range is wide, and it is still too narrow

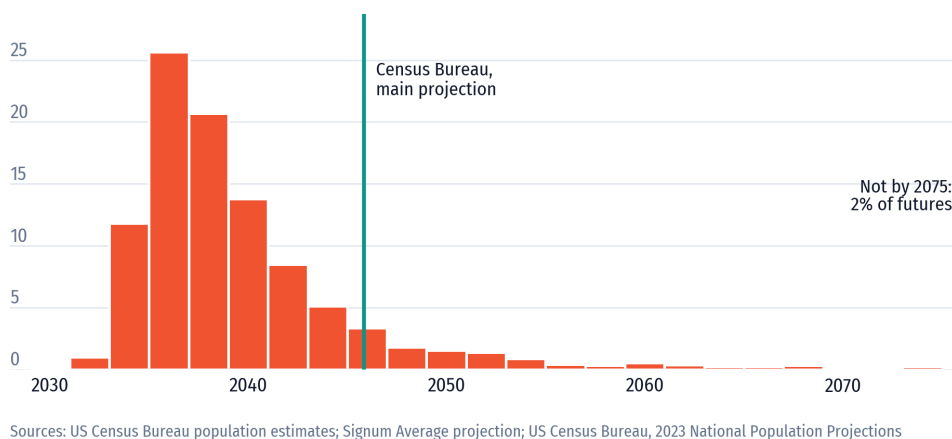
United States: non-Hispanic white share of the population, %
Bands hold 50%, 80% and 95% of simulated futures. The Census Bureau's line uses its older race base



The share most likely falls below half in 2038. Four in five futures put that year between 2035 and 2047; 92% put it before 2050 and 98% before 2075. The Census Bureau's 2023 projection crosses in 2046, near the late edge of our range. It is not a like-for-like comparison: the Bureau's projection starts from older estimates that put the 2025 share at 57.8%, against 56.1% in the estimates it publishes now. Part of that gap is the new race base and part is that immigration in 2022-24 ran far above what the projection assumed.

Most likely the late 2030s

United States: year the non-Hispanic white share falls below half, % of 4,000 simulated futures in each two-year span





The border is not the main cause

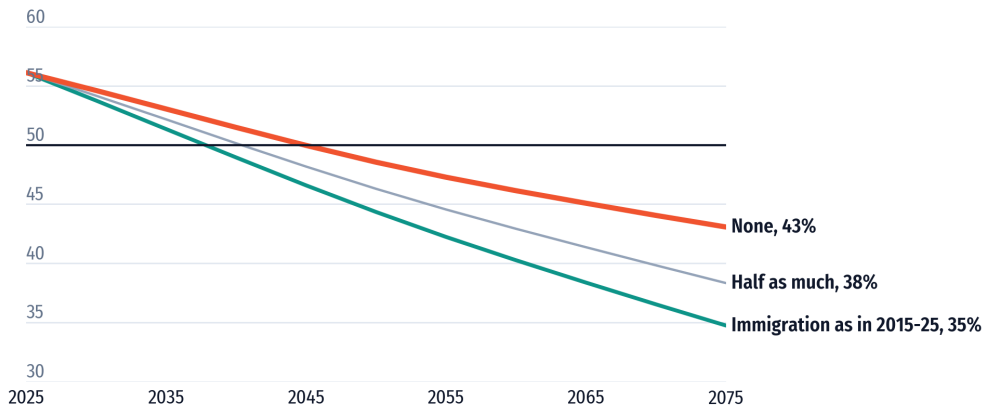
The replacement theory puts immigration at the centre, so we ran the projection with immigration switched off. The state projection cannot do this, because state births and deaths by race are not in the open data and both are folded into how each cohort changes. A national run can: the Census Bureau publishes net migration from abroad by race and Hispanic origin, and we subtract it from each cohort's change.

If net immigration had stopped in 2025 and never resumed, the non-Hispanic white share would still fall below half, in 2045 instead of 2038. By 2075 it would be 43%, eight points higher than with immigration as in 2015-25, and still falling. About three-fifths of the fall (61%) happens with the border shut. With immigration at half its recent pace the line is crossed in 2040.



Shut the border and the share still falls

United States: non-Hispanic white share of the population if net immigration from 2025 continues as in 2015-25, halves, or stops, %



Sources: US Census Bureau; Signum Average projection (national, three groups)

With and without immigration United States: non-Hispanic white share, %, by net migration from abroad after 2025 (national model, three groups)

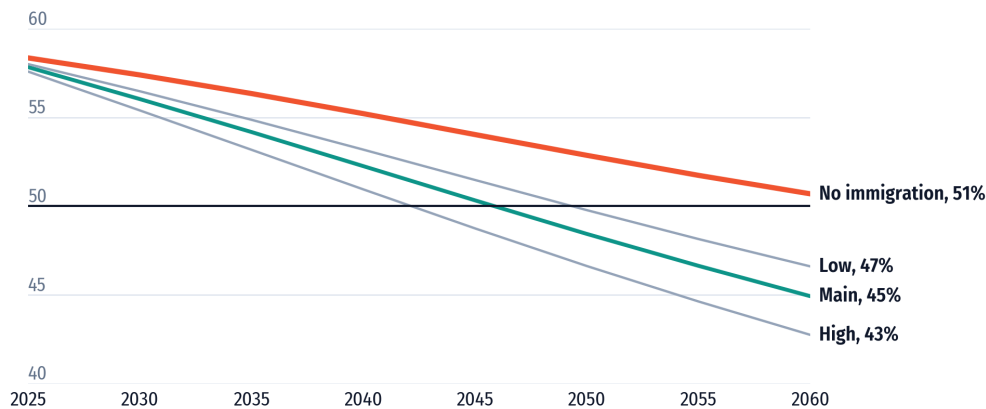
	2050	2075	Below half in	White population 2075, m	Total population 2075, m
As in 2015-25	44.3	34.7	2038	129	373
Half as much	46.3	38.3	2040	125	326
None	48.5	43.1	2045	121	281

Sources: US Census Bureau; Signum Average

The Census Bureau's own projections agree. Its no-immigration scenario, built on older estimates and stricter than ours (no arrivals, while people keep leaving), has the share falling from 58.4% in 2025 to 50.7% in 2060, against 44.9% in its main scenario: again about three-fifths of the fall.

Shut the border and the share still falls

United States: non-Hispanic white share of the population under the Census Bureau's four immigration scenarios, %



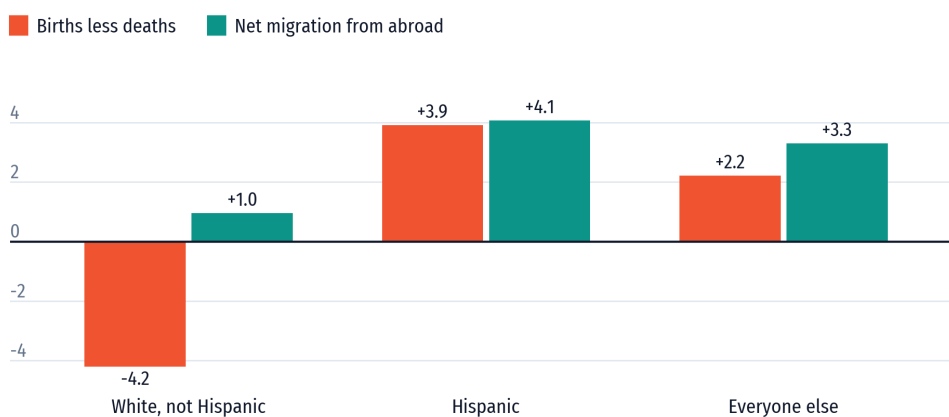
Source: US Census Bureau, 2023 National Population Projections

Why the share falls anyway

The reason is age. White Americans are older than everyone else: they are 73% of people aged 65 and over, and 44% of children under five. Between April 2020 and July 2025, by the Bureau's count, 12.7m non-Hispanic white Americans died and 8.5m were born, a gap of 4.2m. Migration from abroad added 1.0m. In the latest year the gap was 736,000 and migration added 154,000. Every other group, taken together, had more births than deaths.

More deaths than births

United States, April 2020 to July 2025: births less deaths, and net migration from abroad, millions



Source: US Census Bureau, components of population change by race and Hispanic origin (Vintage 2025)

Who arrives

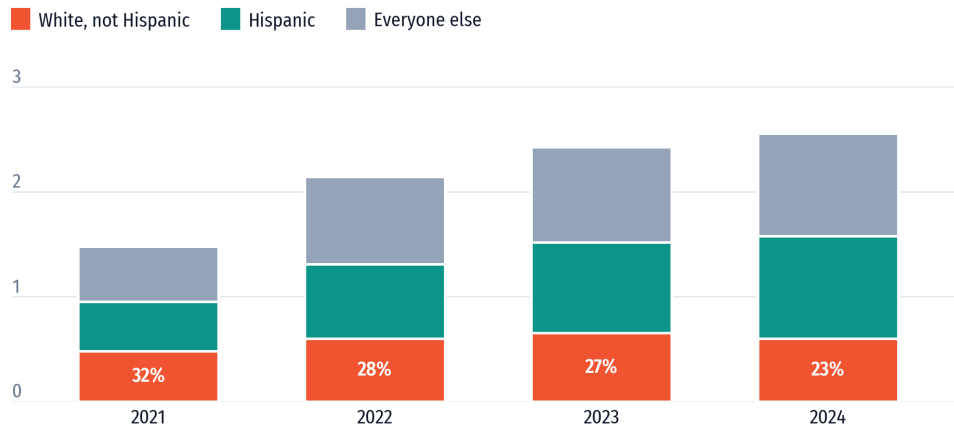
That immigrants are mostly not white is not in dispute. Of the 8.3m net migrants from abroad between April 2020 and July 2025, 12% were non-Hispanic white, 49% Hispanic and 40% everyone else. In 2010-19 the white share was 13%.

For states there is no net figure by race, only arrivals. Of the 2.1m people a year who moved to America from abroad in 2021-24, by the Bureau's survey, 27% were non-Hispanic white, against 56% of residents; 35% were Hispanic, 21% Asian and 11% Black. The white share is higher than in the net figure because arrivals include

Americans coming home. It fell from 32% in 2021 to 23% in 2024 as arrivals rose from 1.5m to 2.6m.

A quarter of arrivals are white

United States: people who lived abroad a year earlier, millions, by group
Labels show the non-Hispanic white share of each year's arrivals



Source: US Census Bureau, American Community Survey (one-year estimates). Includes Americans returning from abroad

The mix varies a great deal by state. Non-Hispanic white people are 18% of arrivals in Texas and New Jersey and over 60% in Montana, Vermont and West Virginia, where few people arrive at all: 1.6 per 1,000 residents a year in West Virginia, against 10.8 in Florida. Of the 50m foreign-born people living in America in 2024, 16% were non-Hispanic white and 45% Hispanic; 4% of white Americans were born abroad.

Who arrives People who lived abroad a year earlier, yearly average 2021-24; foreign-born residents, 2024

	Arrivals a year, 000	per 1,000 residents	White share of arrivals, %	Hispanic share, %	White share of residents, %	Foreign-born, % of residents	White share of foreign-born, %
United States	2,148	6.3	27	35	56	15	16
Texas	252	8.0	18	52	38	18	8
New Jersey	80	8.4	18	40	50	25	17
Delaware	6	5.3	19	22	57	12	14
Georgia	61	5.5	20	33	48	12	11
Rhode Island	9	7.7	21	41	66	16	21
Louisiana	21	4.6	22	38	55	5	11
New Mexico	10	4.7	22	57	35	10	9
Florida	252	10.8	23	53	49	23	13
Maryland	44	7.1	23	25	45	17	11
California	279	7.1	25	32	33	28	13
Mississippi	13	4.3	26	28	55	3	14
Nevada	21	6.4	26	46	43	20	11
Connecticut	26	7.0	27	36	60	16	25
Arizona	50	6.6	27	41	51	13	16
North Carolina	63	5.7	28	32	59	10	14
Arkansas	9	2.8	28	37	67	6	10
Massachusetts	66	9.2	28	22	65	19	24
Oklahoma	17	4.1	28	38	61	7	11
New York	141	7.1	28	29	52	23	19
Indiana	29	4.3	29	26	74	7	12
Minnesota	25	4.4	29	21	75	9	14
Illinois	80	6.3	30	33	57	15	24
Nebraska	12	5.8	31	30	74	9	11
Kentucky	20	4.4	31	36	80	5	16
Hawaii	13	8.9	31	13	21	19	8
Washington	64	8.1	32	20	62	16	20
Virginia	60	6.8	32	24	57	14	16
Iowa	13	3.9	32	26	81	6	15
Pennsylvania	61	4.7	33	26	72	8	22
South Dakota	3	3.7	33	29	79	4	18
Utah	22	6.2	34	44	73	10	14
Alabama	17	3.3	34	26	63	4	15
Colorado	34	5.8	35	42	64	11	19
Ohio	50	4.2	35	18	75	6	25
Wisconsin	20	3.3	36	29	78	6	19

Tennessee	30	4.1	36	33	70	6	17
Kansas	16	5.3	36	34	72	8	11
South Carolina	23	4.3	37	30	62	6	23
Missouri	22	3.6	38	17	76	5	24
Oregon	20	4.7	38	30	70	10	21
Michigan	46	4.5	39	17	72	8	38
Maine	5	3.5	39	9	89	5	40
District of Columbia	10	14.9	39	23	36	16	20
North Dakota	5	6.3	43	18	80	5	17
Idaho	8	4.0	43	34	78	6	20
Wyoming	2	3.3	49	20	80	4	29
Alaska	5	6.5	51	20	57	8	22
New Hampshire	5	3.2	53	10	86	6	38
West Virginia	3	1.6	62	15	89	2	24
Vermont	2	3.8	62	10	89	4	42
Montana	3	3.0	66	12	83	2	48

White: white alone, not Hispanic. Arrivals include Americans returning from abroad. Source: US Census Bureau, American Community Survey

So immigration does lower the white share. It is the smaller part of the story.

What it would take

If not the border, then births? We asked the projection a narrow question: if nothing else changed, how many children would white women need to have for the non-Hispanic white population of 2075 to match that of 2025?

The conditions matter, so here they are in full. White children per woman are multiplied by one factor from 2025 and held there for fifty years. Deaths, immigration and moves between states continue as in the central projection, which averages 2015-20 and 2020-25. Births of mixed-race children are unchanged. The target is the 2075 population, not every year between. "Children per woman" is a total fertility rate: the rate that, with the age pattern of births in the Census Bureau's 2023 projection, yields a state's non-Hispanic white infants from its non-Hispanic white women at each age. Nationally it is 1.40, and it reproduces the Bureau's count of white births to within 0.3%. It counts only children classed as white alone and not Hispanic, so it is below the official rate for white mothers, some of whose children are classed as mixed or Hispanic.

The country as a whole

One factor is applied in every state, and the target is the national total. As things are, the white population falls by 28%, from 192m to 138m. Holding it level takes a rise of 56% in white births per woman: about 2.2 children each, against 1.4 today. Holding the white share at 56%, not just the number, takes two and a half times today's rate, about 3.5 children, which America last saw during the baby boom.

The number needed to hold the population is close to the textbook replacement rate of 2.1 because white immigration is small. The number needed to hold the share is far higher because everyone else is growing: new babies take 25 years to become parents, and the large older white generations die in the meantime.

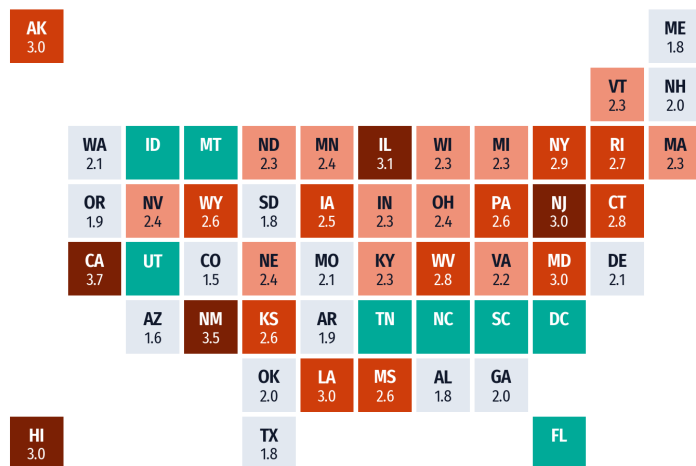
Each state on its own

The map and table ask a different question: for each state separately, how many children would white women in that state need for that state's white population to hold? Here moves between states count, because a state must replace those who leave as well as those who die.

What it would take, state by state

Each state on its own: children per white woman needed there, from now on, for that state's non-Hispanic white population in 2075 to match 2025's. The country as a whole needs 2.2

■ Grows anyway
 ■ Under 2.1
 ■ 2.1 to 2.5
 ■ 2.5 to 3
 ■ 3 or more



Sources: US Census Bureau population estimates; Signum Average projection. Only births change; deaths, immigration and moves between states continue as in 2015-25. State figures do not add up to the national one

In seven states and Washington, DC, the white population grows with no change in births, because white Americans from elsewhere keep arriving: Idaho (up 39% by 2075), South Carolina, Tennessee, Montana, Utah, North Carolina and Florida. Where they leave, the arithmetic is harsh: California would need 3.7 children per white woman, New Mexico 3.5 and Illinois 3.1.

What it would take Non-Hispanic white population, and the children per white woman needed to hold it, or its share, at the 2025 level in 2075

	White population 2025, m	Change by 2075 as things are, %	Children per woman now	Needed to hold the population	Needed to hold the share
United States	191.8	-28	1.4	2.2	3.5
District of Columbia	0.3	+28	0.7	fewer than now	0.7
Idaho	1.6	+38	1.6	fewer than now	4.3
South Carolina	3.4	+29	1.4	fewer than now	2.5
Montana	1.0	+10	1.5	fewer than now	3.6
Tennessee	5.1	+10	1.5	fewer than now	3.2
Utah	2.6	+10	1.5	fewer than now	4.4
North Carolina	6.6	+3	1.4	fewer than now	3.4
Florida	11.7	+1	1.4	fewer than now	3.6
Colorado	3.8	-16	1.1	1.5	3.5
Arizona	3.9	-8	1.4	1.6	3.7
Texas	11.8	-13	1.4	1.8	3.6
South Dakota	0.7	-4	1.7	1.8	5.4
Maine	1.3	-17	1.3	1.8	5.2
Alabama	3.2	-11	1.6	1.8	2.7
Oregon	3.0	-35	0.9	1.9	4.1
Arkansas	2.1	-13	1.6	1.9	3.2
New Hampshire	1.2	-22	1.3	2.0	4.6
Georgia	5.4	-22	1.5	2.0	3.5
Oklahoma	2.4	-28	1.3	2.0	3.8
Washington	4.9	-31	1.2	2.1	4.7
Missouri	4.7	-20	1.6	2.1	3.3
Delaware	0.6	-27	1.3	2.1	3.9
Virginia	5.1	-32	1.4	2.2	3.4
Indiana	5.1	-27	1.6	2.3	4.4
Kentucky	3.7	-24	1.7	2.3	4.5
Massachusetts	4.7	-44	1.1	2.3	4.3
Vermont	0.6	-33	1.3	2.3	4.1
North Dakota	0.6	-26	1.5	2.3	5.9
Wisconsin	4.6	-32	1.4	2.3	4.0
Michigan	7.3	-34	1.4	2.3	3.1
Nebraska	1.5	-30	1.6	2.4	5.1
Ohio	8.9	-35	1.5	2.4	4.0
Minnesota	4.3	-34	1.4	2.4	4.9
Nevada	1.4	-42	0.9	2.4	4.9
Iowa	2.6	-31	1.6	2.5	5.4

Mississippi	1.6	-35	1.6	2.6	2.5
Pennsylvania	9.4	-43	1.3	2.6	4.3
Kansas	2.1	-33	1.6	2.6	3.8
Wyoming	0.5	-37	1.5	2.6	4.1
Rhode Island	0.7	-50	1.0	2.7	4.6
Connecticut	2.2	-49	1.2	2.8	4.7
West Virginia	1.6	-43	1.6	2.8	4.3
New York	10.3	-50	1.4	2.9	2.8
Maryland	2.8	-51	1.4	3.0	3.9
Louisiana	2.5	-45	1.6	3.0	3.2
Alaska	0.4	-52	1.2	3.0	3.9
Hawaii	0.3	-44	1.2	3.0	2.1
New Jersey	4.7	-46	1.6	3.0	4.2
Illinois	7.2	-55	1.4	3.1	3.3
New Mexico	0.7	-58	1.2	3.5	3.6
California	12.8	-65	1.2	3.7	3.9

Each state is solved on its own; the United States row applies one factor everywhere. Children per woman: a total fertility rate counting only children classed as white alone and not Hispanic. Sources: US Census Bureau; Signum Average

The state numbers are not a recipe for the national one, and they do not add up to it. Nationally, moves between states cancel out: a family that leaves California for Idaho lowers one state's count and raises another's. The country needs 2.2 overall however the births are spread, and no state needs to reach its own figure for the national total to hold. The state figures also carry the moves of 2015-25 forward for fifty years, the least certain part of the projection; they show what the recent pattern implies, not what will happen.

Other levers

Births are one lever of several, and we solved for it alone. The fall of 54m over fifty years could also be met by about 1.1m more white immigrants a year; net white immigration has run at about 180,000 a year since 2020. That is arithmetic, not a projection: it ignores the children those immigrants would have. Longer lives would slow the fall. And the count itself can move without anyone being born or arriving: mixed families identifying as white, or a wider definition, which the next section shows is the largest lever of all.

Who counts as white

The definition matters more than any assumption about migration. The narrow count, the one behind every figure above, includes only people who tick white, nothing else, and are not Hispanic. A child with one white parent and one Asian parent is not in it.

Three definitions of white United States: share of the population, %; median and 80% range

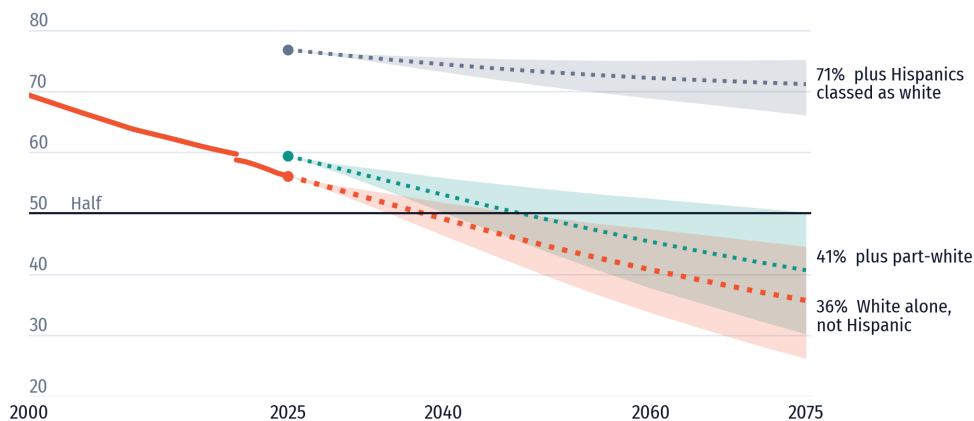
	2025	2040	2050	2060	2075
White alone, not Hispanic	56	49 (46-52)	45 (40-49)	41 (34-47)	36 (26-45)
plus part-white, not Hispanic	59	53 (50-56)	49 (44-54)	45 (38-52)	41 (30-50)
plus Hispanics classed as white	77	74 (73-76)	73 (71-75)	72 (69-75)	71 (66-75)

Sources: US Census Bureau; Signum Average

Count instead everyone who is non-Hispanic and white or part-white, and the 2075 median is 41%; there is a 91% chance that this share too is below half by then, most likely from 2047. Add Hispanics whom the Bureau's estimates class as white, alone or mixed, and the figure is 71%, down from 77% today. On that count the white majority holds in all but six of 4,000 simulated futures.

One forecast, three answers

United States: share of the population counted as white under three definitions, %
Dotted lines are the median projection; bands hold 80% of 4,000 simulated futures



Sources: US Census Bureau population estimates; Signum Average projection

The Bureau changed its race base in 2020 (small step in the line). Hispanics are classed by race as in its 2025 estimates

Neither wider count is right either. The middle one holds the part-white share of mixed-race Americans at its 2025 level (92%), and counts mixed-race births per woman of any group at their 2025 rate, when intermarriage has been rising. The broad one rests on the Bureau's "modified race" procedure, which reassigns people who chose "some other race" in the census: its estimates class about four in five Hispanics as white alone, but in the 2020 census only about a fifth of Hispanics chose that for themselves. How people see themselves lies somewhere between the lines, and it moves. Italians, Irish and Jews were once counted apart from America's white majority; their grandchildren are not.

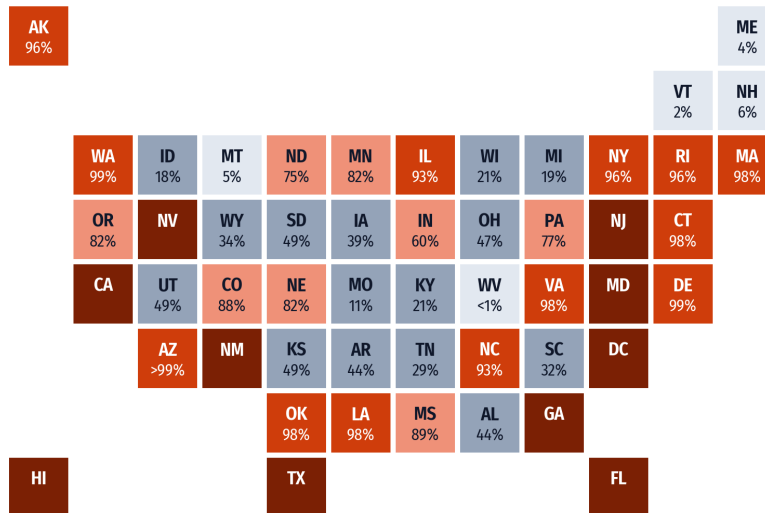
Fifty different countries

Nine states and Washington, DC, already have no white majority on the narrow count: Hawaii, California, New Mexico, Texas, Nevada, Maryland, Georgia, New Jersey and Florida. Fifteen more are likelier than not to join them by 2050, and six more by 2075.

20 states will probably keep a white majority

Chance that the non-Hispanic white share of each state is below half by 2075

■ Already ■ Over 90% ■ 50-90% ■ 10-50% ■ Under 10%



Sources: US Census Bureau population estimates; Signum Average projection

That leaves 20 states that will probably keep a white majority to 2075. For five of them (Maine, Vermont, New Hampshire, Montana and West Virginia) the chance of losing it is under one in ten. The ranges for single states are wide, and widest for small ones: North Dakota's 80% range for 2075 runs from 18% to 61%.

Every state Non-Hispanic white share of the population, %, and the chance it is below half

	2025	2050 (80% range)	2075 (80% range)	Below half by 2050	by 2075	Median year
Hawaii	21	18 (14-24)	18 (10-29)	already	already	already
California	33	22 (17-27)	14 (8-23)	already	already	already
New Mexico	35	24 (19-30)	17 (9-26)	already	already	already
Texas	37	26 (22-31)	20 (13-28)	already	already	already
District of Columbia	38	37 (29-45)	35 (23-50)	already	already	already
Nevada	42	26 (21-32)	17 (10-25)	already	already	already
Maryland	45	32 (28-36)	22 (15-29)	already	already	already
Georgia	48	37 (33-40)	28 (22-35)	already	already	already
New Jersey	50	37 (31-42)	27 (18-37)	already	already	already
Florida	50	39 (32-46)	32 (22-44)	already	already	already
Arizona	51	39 (33-45)	31 (21-42)	over 99%	over 99%	2027
New York	52	43 (36-50)	35 (24-49)	93%	96%	2030
Mississippi	55	49 (45-53)	43 (34-51)	66%	89%	2045
Louisiana	55	46 (41-50)	37 (28-46)	89%	98%	2039
Delaware	57	44 (39-49)	33 (24-42)	94%	99%	2038
Illinois	57	47 (40-53)	37 (25-49)	77%	93%	2042
Alaska	57	44 (36-52)	30 (18-44)	86%	96%	2039
Virginia	57	45 (40-50)	36 (26-44)	90%	98%	2040
North Carolina	59	48 (44-53)	40 (31-49)	68%	93%	2046
Oklahoma	59	46 (40-51)	34 (24-44)	86%	98%	2042
Washington	61	43 (37-50)	30 (20-41)	91%	99%	2040
Connecticut	61	43 (36-51)	30 (19-42)	87%	98%	2041
South Carolina	62	58 (53-62)	53 (43-62)	2%	32%	after 2075
Alabama	62	57 (53-60)	51 (41-58)	2%	44%	after 2075
Colorado	63	50 (43-56)	39 (26-51)	52%	88%	2049
Massachusetts	66	49 (43-54)	34 (24-44)	62%	98%	2048
Rhode Island	66	49 (42-55)	33 (21-46)	60%	96%	2048
Arkansas	68	59 (54-64)	51 (40-61)	2%	44%	after 2075
Oregon	70	54 (47-61)	40 (26-54)	23%	82%	2058
Tennessee	70	62 (57-67)	55 (43-65)	under 1%	29%	after 2075
Kansas	71	61 (54-66)	50 (36-62)	2%	49%	after 2075
Pennsylvania	72	58 (52-63)	44 (32-55)	4%	77%	2064
Michigan	72	64 (60-68)	56 (46-64)	under 1%	19%	after 2075
Utah	73	61 (54-68)	50 (36-64)	3%	49%	after 2075
Nebraska	73	57 (51-63)	42 (30-54)	7%	82%	2061
Indiana	73	60 (56-65)	48 (38-57)	under 1%	60%	2071
Minnesota	74	59 (54-63)	44 (35-52)	1%	82%	2065

Ohio	75	63 (59-66)	50 (41-58)	under 1%	47%	after 2075
Missouri	75	68 (64-72)	60 (50-68)	under 1%	11%	after 2075
Wisconsin	78	68 (63-72)	57 (45-67)	under 1%	21%	after 2075
Idaho	78	69 (61-76)	62 (44-76)	under 1%	18%	after 2075
South Dakota	78	65 (57-72)	51 (34-66)	1%	49%	after 2075
Kentucky	80	68 (64-73)	57 (46-66)	under 1%	21%	after 2075
North Dakota	80	59 (46-71)	38 (18-60)	17%	75%	2061
Iowa	81	67 (61-72)	52 (40-63)	under 1%	39%	after 2075
Wyoming	81	71 (59-81)	59 (33-80)	2%	34%	after 2075
Montana	83	76 (70-81)	70 (55-80)	under 1%	5%	after 2075
New Hampshire	87	77 (72-82)	67 (54-77)	under 1%	6%	after 2075
West Virginia	89	82 (78-85)	74 (63-82)	under 1%	under 1%	after 2075
Maine	89	80 (74-85)	70 (56-81)	under 1%	4%	after 2075
Vermont	89	83 (76-88)	76 (62-86)	under 1%	2%	after 2075

Sources: US Census Bureau; Signum Average

How far to trust it

Not as far as the decimal points suggest. Four tests bear on it, and one of them went badly.

The method over ten years

The projection method was fixed in writing before it was run, and four versions were tried on the past. The version used beat both naive guesses over ten years, but only narrowly against a simple continuation of the trend, and over five years the trend was slightly better. It put the national white share in 2020 1.3 points too low.

Would it have called 2020? Average miss across 51 states in the non-Hispanic white share, percentage points

	From 2010, five years on	From 2010, ten years on	From 2015, five years on
Ratios only; mixed-race births per mixed-race woman	0.88	2.00	0.36
Ratios only; mixed-race births per woman of any group	0.84	1.84	0.33
Ratios and differences; per mixed-race woman	0.67	1.27	0.25
Ratios and differences; per woman of any group (used)	0.64	1.18	0.25
Naive: shares stay as at the start	1.81	3.56	1.75
Naive: the trend continues	0.67	1.28	0.23

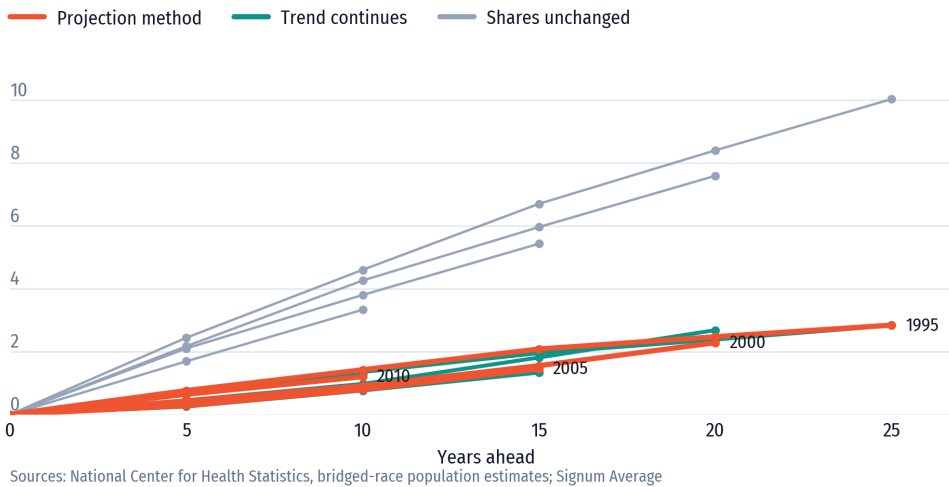
Sources: US Census Bureau; Signum Average

The method over 25 years

Ten years is a short test for a fifty-year projection. The health-statistics agency publishes population estimates for 1990 to 2020 in consistent race categories (four races, with mixed-race people assigned to one of them, so not the categories used elsewhere in this report). On those, the method was started in 1995, 2000, 2005 and 2010 and run to 2020.

As good as the trend, no better

Average miss across 51 states in the non-Hispanic white share, percentage points, by years ahead
Four starting years (labelled), each run to 2020, on 1990-2020 data with consistent race categories



After 20 to 25 years it missed each state's white share by 2.3 to 2.8 points on average. Assuming no change missed by 7.6 to 10. A straight line through the trend missed by 2.7 to 2.8. The method beat the trend from 2000 and 2010, tied it from 1995 and lost to it from 2005. It is as good as the trend and no better; what it adds is detail by age, the scenarios and the what-ifs.

Twenty-five years of hindsight Each start year run to 2020: average miss across 51 states in the non-Hispanic white share, percentage points

	Years ahead	Shares unchanged	Trend continues	Projection method	Projection's national error
From 1995	25	10.03	2.83	2.84	-0.4
From 2000	20	7.59	2.68	2.28	-2.7
From 2005	15	5.43	1.33	1.45	-1.8
From 2010	10	3.33	1.26	1.20	-1.3

National error: projected less actual white share of the United States in 2020, percentage points. Sources: National Center for Health Statistics, bridged-race population estimates; Signum Average

Its errors ran one way. At three of the four starts it put the national white share too low, by 1.3 to 2.7 points, and the white share fell more slowly than projected in 40 to 47 of the 51 states. Immigration and Hispanic birth rates both fell after 2007, which no method fed on the 1990s or early 2000s could know. We have not shifted the projection to allow for this, because one period cannot say whether the same will happen again. But it is a reason, with the Census Bureau's later date, to think the fall is more likely to be slower than our median than faster.

The ranges

The ranges were tested twice, with different results.

First, an odds model of the same design was fitted to the 1990s only and launched in 2000. It was too confident. In 2010, 39 of 51 states fell inside their 80% range; in 2020, 31 did, and the national share (60.9%) fell outside the 95% range (56.7% to 60.0%). The 1990s were a calm decade for the series the model watches, so it expected calm.

Second, the ranges actually published here were set against the errors the method actually made. The published model is fitted to 2010-25, which was not calm, and its ranges are about 60% wider at the same

number of years ahead. Had ranges of that width been placed around the four past projections, they would have held 40, 48, 44 and 47 of the 51 states at 10, 15, 20 and 25 years, where 41 is the target, and all 14 of the national errors.

Do the published ranges hold the method's past errors? States whose actual 2020 share falls inside a range of the published width, placed around each past projection

	Years ahead	Inside the 80% range (target 41)	Inside the 95% range (target 48)	Typical state 80% range, points	States projected too low
From 2010	10	40 of 51	51 of 51	3.5	40
From 2005	15	48 of 51	51 of 51	5.8	47
From 2000	20	44 of 51	51 of 51	7.9	45
From 1995	25	47 of 51	49 of 51	10.1	15

Sources: National Center for Health Statistics, bridged-race population estimates; Signum Average

We publish ranges stretched by the most the second test called for, which is 8%. That is a choice, made with both results in view, and the other choice is defensible: stretch the ranges by what the first test called for, on the ground that 2010-25 may prove as poor a guide to the next decades as the 1990s were to the 2000s. On that harsh reading the median is unchanged but the chance of the white share falling below half by 2050 is 76%, not 92%; by 2075 it is 88%, not 98%; and the 80% range for 2075 runs from 14% to 56%, not 26% to 45%.

Two readings of the same forecast United States, non-Hispanic white share: the median is the same (45% in 2050, 36% in 2075); the ranges differ

	Below half by 2050	Below half by 2075	Year it falls below half, 80% range	Share in 2050, 80% range	Share in 2075, 80% range
Published	92%	98%	2035 to 2047	40 to 49	26 to 45
Harsh reading	76%	88%	2034 to after 2075	33 to 55	14 to 56
Model before any stretch	94%	99%	2035 to 2046	40 to 49	27 to 44

Sources: US Census Bureau; Signum Average

The scenarios

The scenarios without probabilities are a last check. Carrying forward the migration and deaths of 2015-20 gives a 2075 share of 38%; those of 2020-25 give 34%. Both lie well inside the 80% range, which is as it should be: two recent five-year periods should not span everything that can happen in fifty years.

The scenarios, without probabilities United States: non-Hispanic white share, %, and total population

	2050	2075	Below half in	Population in 2075, m
Low migration (as in 2015-20)	46.1	38.3	2040	359
Central	44.7	35.8	2038	386
High migration (as in 2020-25)	43.3	33.8	2037	415
Central, birth rates converge	45.6	38.5		378

Sources: US Census Bureau; Signum Average

What this analysis cannot tell you

Whether the next fifty years will vary as 2010-25 did. That period includes a pandemic, an immigration surge and its collapse, but not a different immigration regime, a war or a larger shock. Nothing here was tested beyond 25 years.

The projection cannot foresee people changing how they describe themselves, or the government changing the boxes. A 2024 federal standard adds a Middle Eastern or North African category, which will move people out of "white" by rule, and merges the race and Hispanic-origin questions, which will change how Hispanics are counted.

It treats a shock as hitting every age in a group alike, holds each state's birth rates in a fixed relation to the nation's, and treats states as independent of one another once the national swing is taken out.

It says nothing about anyone's intent.

Appendix: data and methods

Data

US Census Bureau population estimates by state, sex, age, race and Hispanic origin: the intercensal estimates for 2000-2010, Vintage 2020 for 2010-2020 and Vintage 2025 for 2020-2025. The national benchmark and the no-immigration scenario are the Bureau's 2023 National Population Projections. Vintage 2025 uses a newer race base than earlier vintages: it puts the non-Hispanic white share in 2020 at 58.8%, against 59.7% in Vintage 2020. No change is ever measured across vintages, and the history chart shows the step. The estimates use the Bureau's "modified race" categories, which reassign people who chose "some other race" in the census; this is why they class most Hispanics as white.

Projection

A Hamilton-Perry cohort projection in five-year steps from 2025, for 51 areas, two sexes, 18 age groups and six groups (non-Hispanic white, Black, Native American, Asian or Pacific Islander, and two or more races; and Hispanic of any race). For each cell the method takes the change in the size of a cohort over a past five-year period, which combines deaths and migration, and applies it again. Where a cohort shrank its ratio is applied; where it grew the absolute increase is added, which stops small, fast-growing groups from exploding (Hauer, 2019). Births are children under five per woman aged 15-49 in 2025; children of two or more races are counted per woman of any group. Cells of fewer than 50 people borrow the national figure. The central projection averages cohort change in 2015-20 and in 2020-25. The specification was written before the model was run.

The two wider definitions of white apply 2025 shares, by state and age, to the projected groups: 92% of the non-Hispanic two-or-more group are part-white, and 85% of Hispanics are classed as white alone or mixed.

Odds

A Bayesian model, fitted with PyMC, describes how much the projection's inputs moved from year to year in 2010-25: for each group, yearly cohort change net of age structure (which moves with deaths and migration) and infants per woman; and for each state, its departure from the national figure. Cohort change is modelled as reverting to a long-run level with correlated shocks across groups; birth rates as a random walk with no drift. The projection is then run 4,000 times, each under one draw of the model's parameters and one simulated future. The ranges quoted hold 80% of those futures, after the small stretch described under Tests. Yearly swings were about eight times larger for Hispanic and Asian Americans than for white Americans, which is migration.

Tests

Short test. Started in 2010 with cohort change from 2005-10, the projection missed states' non-Hispanic white share in 2020 by 1.18 percentage points on average. Assuming shares stayed as in 2010 missed by 3.56; assuming each share kept changing at its 2005-10 pace missed by 1.28. Nationally the method put the 2020 share 1.3 points too low.

Long test. On the health-statistics agency's 1990-2020 estimates, which keep race categories consistent (and have no mixed-race group), the method was started in 1995, 2000, 2005 and 2010 and run to 2020. It missed states by 2.84, 2.28, 1.45 and 1.20 points on average; a straight-line trend missed by 2.83, 2.68, 1.33 and 1.26; no change by 10.0, 7.6, 5.4 and 3.3. So the method is as good as the trend and no better. At three of the four starts it put the national white share too low, by 1.3 to 2.7 points.

Tests of the ranges. An odds model fitted to the 1990s only and launched in 2000 was too narrow: 39 of 51 states fell inside their 80% range in 2010 and 31 in 2020, and the national figure fell outside its 95% range. Its ranges were tight because its decade was calm. The published model's ranges, fitted to 2010-25, are about 60% wider at the same horizon. Set against the

method's actual errors from the four starts, state by state, they would have held 40, 48, 44 and 47 of 51 states at 10, 15, 20 and 25 years, and all 14 national errors. The published ranges are stretched by the most that this second test called for (8%). Stretching them by what the first test called for gives the harsh reading in the box above: a 76% chance of falling below half by 2050, and a 2075 range of 14% to 56%.

No immigration

A separate national run of the same cohort method on the 2025 base, in three groups (non-Hispanic white, Hispanic, everyone else). Net migrants from abroad by group are the Census Bureau's own (its components of population change for 2020-25; for 2015-20 the total, split by group as in 2010-19), given the age and sex pattern of its 2023 projection, and are subtracted from each cohort's change. "None" means zero net migration from abroad after 2025. The Bureau's no-immigration scenario is stricter: no arrivals while people keep leaving. With migration left in, this simpler model puts the white share in 2075 at 34.7%, a point below the main state-level projection.

Who arrives

Net migration by group is the Bureau's, national only. State figures are from the American Community Survey's one-year tables for 2021-24: people aged one and over whose home a year earlier was abroad, by race and Hispanic origin. That is a gross count that includes Americans returning from abroad, which is why white people are 27% of it but 12% of net migration. State figures pool four years because single years are noisy.

Births needed

A what-if on the central projection: white children per woman are multiplied by one factor, from 2025 on, until the white population (or share) of 2075 equals that of 2025, with deaths, immigration, moves between states and mixed-race births left as they are. The national figure applies one factor everywhere; each state figure is solved for that state alone. Children per woman is a total fertility rate: the rate that, with the age pattern of the Bureau's 2023 projection, yields a state's non-Hispanic white infants from its non-Hispanic white women by single year of age. Nationally it reproduces the Bureau's count of white births to within 0.3%. It counts only children classed as white alone and not Hispanic, so it is below the official rate for white mothers, some of whose children are classed as mixed or Hispanic.

Not modelled

Changes in how people describe themselves, changes to official categories, rising intermarriage, and shocks that hit some ages more than others. State-level births and deaths by race are folded into the cohort ratios, so the split between ageing and migration is taken from the Census Bureau's national scenarios, not from our model.

The data behind every chart can be downloaded from the data on GitHub, free to reuse with credit.

The yearly indexes behind the odds

The cohort index for a group in a year is the log of the people aged one and over a year later, over the number expected if each age had changed by its average for 2010-2025. It is computed from single years of age, always inside one vintage of estimates, giving 15 yearly values per group. Without the adjustment for age the index falls steadily for white Americans simply because they are ageing, which the projection already handles; a first version of the model made that mistake, read ageing as a near-permanent shock and failed its diagnostics. The birth index is the log of infants per woman aged 15-49. State indexes are computed for non-Hispanic white people and for everyone else together, less the national index.

The three models

National cohort index: $a[t] = \mu + x[t]$, $x[t] = \phi x[t-1] + e[t]$, with e multivariate normal across the six groups (LKJ(2) prior on the correlation), $\mu[g] \sim \text{Normal}(\text{sample mean, twice the sample standard deviation})$, $\phi \sim \text{Beta}(2, 2)$, standard deviations $\text{HalfNormal}(0.02)$. Birth index: $b[t] = b[t-1] + n[t]$ with no drift, n multivariate normal across groups, standard deviations $\text{HalfNormal}(0.05)$. State deviation: an AR(1) around a state level, with a standard deviation that depends on the state's population. Each is sampled with NUTS: four chains, 1,500 tuning and 1,000 kept draws each. The final fit had $\hat{r}$ of 1.00 throughout, at least 1,100 effective samples per parameter and one divergent transition in 4,000. The fit to 2010-15 used for the calibration check is weaker, as five data points would suggest: 13 divergent transitions and $\hat{r}$ up to 1.01. Posterior means: persistence 0.91 nationally; yearly standard deviation of the cohort index 0.06 log points for white Americans, 0.54 for Asian and 0.47 for Hispanic Americans.

From model to ranges

In each five-year step of a simulated future, a group's cohort change in the central projection is multiplied by the exponential of the sum of five simulated yearly indexes less five times the index's average over the central projection's base years; the state deviation enters the same way; and children per woman are multiplied by the exponential of the simulated change in the birth index since 2025. The shock is the same at every age in a group, which is a simplification: migration is concentrated among young adults.

Stretching the ranges

For each past launch and each state, the projection's error in 2020 is compared with the published model's range for that state the same number of years ahead. The stretch a test calls for is the factor by which every state's range must be multiplied about its median for 80% of states to fall inside the 80% range, and likewise 95%; the larger is used. The second test called for 1.08 at ten years and less than 1 at 15, 20 and 25, so 1.08 is applied, phased in over the first ten years. The first test (the 1990s model) called for 1.31 at ten years and 1.38 at twenty for states, and 1.39 and 2.48 for the nation, the latter being the stretch that would have put the actual national share at the edge of the range; those give the harsh reading. Each simulated future's distance from the median is multiplied by the factor for its year. Beyond the years tested the factor is held, which is a floor.

What changed along the way

The specification, written before any model was run, records every later change with its reason: the switch from Vintage 2024 to Vintage 2025 estimates, which were already published and should have been used from the start; the age adjustment to the yearly index; a tighter prior on the long-run level; a fertility measure in standard units, and the dropping of a survey-based scaling that failed a check against official births; the long test; and the choice of which test of the ranges sets what is published, which was made after both results were known. The first version, on Vintage 2024 data and without odds, put the national crossing at 2040 and the 2074 share at 37%.

Reproducing the results

The code that downloads every input, runs the backtest, the projection and the simulations and draws the charts is kept in the study's private repository; the data and these methods are public. Random seeds are fixed, so a rerun gives the same numbers to within sampling error across machines. The published numbers come from a single end-to-end run.